

Forecasting Household Appliance Energy Demand

A Comparative Study of Benchmark, Classical, Machine-Learning, and Foundation-Model Forecasters

Ameer Hamza | Student ID:24178176 | [GitHub](#)

Introduction

Accurately forecasting short-term household appliance energy demand is a practical problem for smart-home energy management, demand response, and grid-capacity planning. This report presents an end-to-end 24-hour-ahead forecasting study on the UCI Appliance Energy Prediction dataset (Candanedo, Feldheim, & Deramaix, 2017), which records energy use in a low-energy house in Belgium at 10-minute resolution over roughly 4.5 months, alongside indoor sensor readings and local weather observations.

This study aims more than just to benchmark one model, as it attempts to benchmark four classes of forecasting approach: (1) simple statistical benchmarks, (2) a classical time-series model (SARIMAX), (3) a feature-based machine-learning model (XGBoost), and (4) a pretrained foundation model (Chronos) used in a zero-shot fashion. All four categories are assessed with the same 14 rolling-origin 24-hour test folds and error metrics (RMSE, MAE, MAPE) to eliminate the bias that may occur due to different measurements used to assess performance.

Data and Exploratory Analysis

The data was resampled from 10-minute resolution to hourly resolution (mean aggregation), and no missing data or gaps in the time series. Throughout, the target variable is Appliances, which is the average energy use of appliances per hour, in Wh.

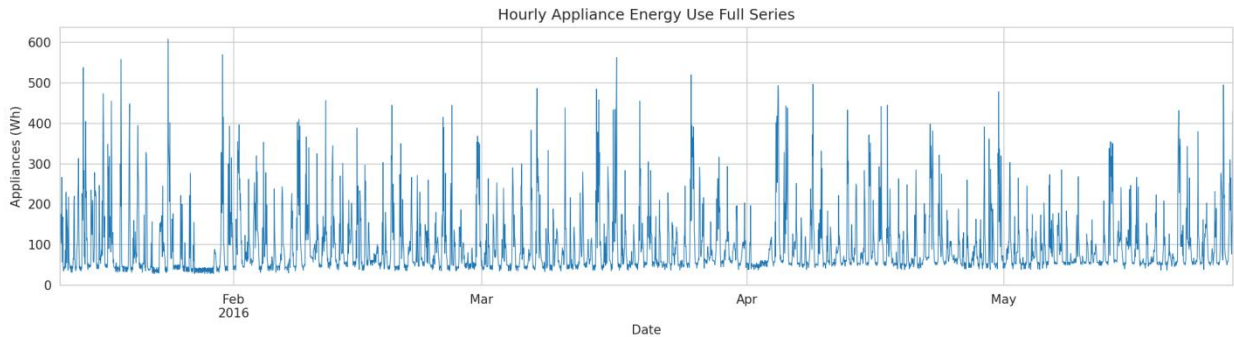


Figure 1: Full hourly Appliances series, Jan–May 2016

The strong daily time of day/week cycle identified by exploratory analysis (STL decomposition, hour-of-day and day-of-week averages, and ACF) showed (Figure 1) that the highest use was during the morning and evening peaks and the lowest use was overnight. A weaker but still present weekly cycle was also observed, which is a difference between the routines of the weekdays and the weekends.

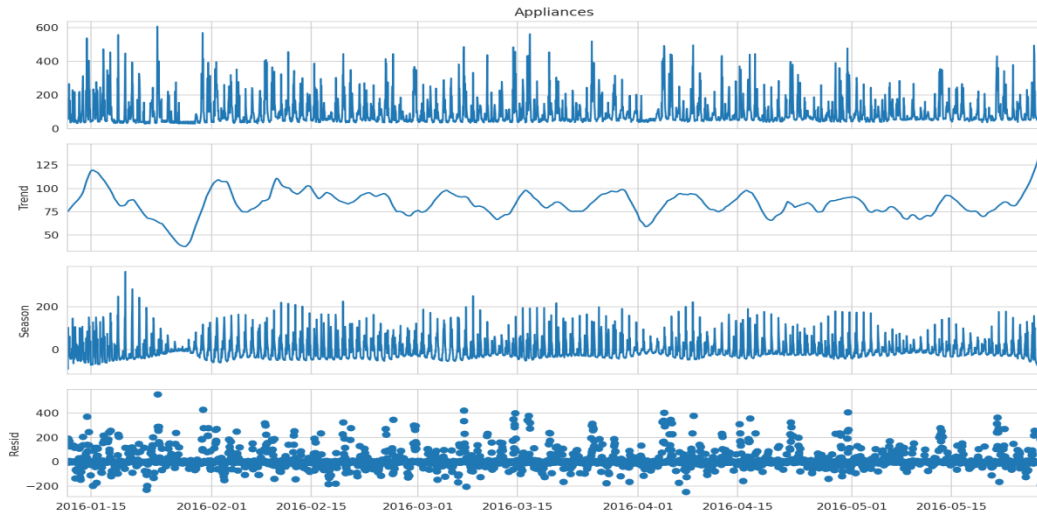


Figure 2: STL decomposition (trend, daily seasonal, residual) of the Appliances series

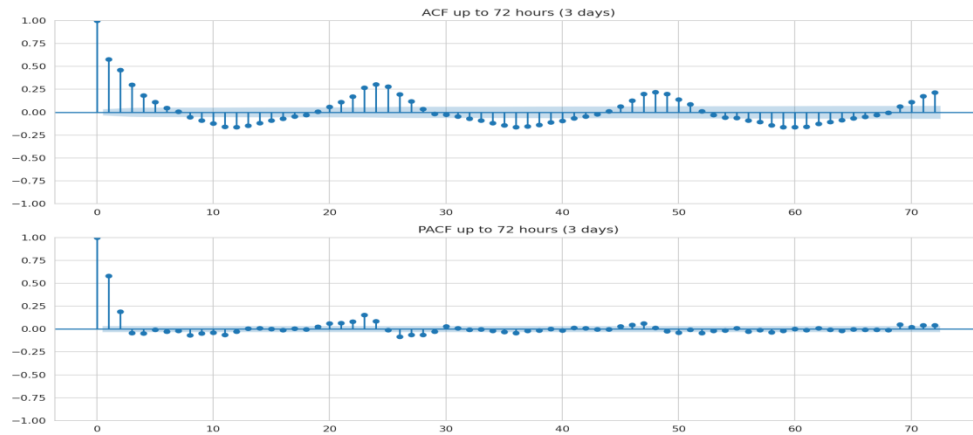


Figure 3: ACF and PACF of Appliances up to 72 hours, showing repeating spikes at multiples of 24 hours

The strong seasonal swing of the series allowed the use of neither trend-differencing nor log-differencing due to its stationarity in levels (as confirmed by both ADF and KPSS tests), although seasonal differencing was explored in the context of SARIMAX because of the clear 24-hour cycle.

Methodology

Forecasting task definition

The task is formulated as: predict Appliances at a 24-hour time horizon, hourly time step, from any time in the past. The last 14 days (336 hours) were reserved for a test period and divided into 14 non-overlapping 24-hour rolling-origin folds, instead of using a one-time training/test split, to prevent the risk of a single chance of luck or bad luck in a single window and, therefore, allow each model its own opportunity to perform and report the result as mean $\pm$ std.

Benchmarks

Five simple benchmarks were added as reference base: Mean (forecast = historical average); Naive (last observed value repeated); Daily Seasonal Naive (repeats the value from 24 hours prior); Weekly Seasonal Naive (repeats the value from 168 hours / 7 days prior); Drift (naive plus a linear trend extrapolated across the training history).

SARIMAX

The EDA suggests a 24-day period ($s = 24$), which is the highest period in the grid search, so this period is used in the seasonal grid search ($p \in [0,6]$, $d \in [0,2]$, $q \in [0,6]$ and $s = 24$; 147 combinations). The final selected order was SARIMAX(5,1,6) $\times$ (0,1,1,24). To avoid recalculating the expensive seasonal maximum-likelihood optimization 14 times, a single reference model was fit and its estimated parameters applied to the current up-to-date history of each fold (a standard, efficient procedure known as rolling forecasts).

XGBoost (feature-based ML model)

A direct multi-horizon design was employed, since training one model to make one step prediction at a time (recursively) would accumulate errors over 24 steps. A single XGBoost regressor was trained on a long-format dataset, where each row contains an origin time t_0 , a horizon $h \in [1,24]$, and a target value v at t_0+h , with the horizon h itself added as a feature. The weather/sensor readings were current (t_0 only), and the calendar features included hour, day-of-week, month, weekend flag, and cyclical sine/cosine encodings, where the latter were computed over the given window of time. The weather/sensor readings were current (t_0 only), and the calendar features were hour, day-of-week, month, weekend flag, and cyclical sine/cosine encodings (computed over the specified window of time). More critically, no future weather values were used as features, as this would not be true in a real deployment where such values would not be truly available at forecast time (see Section 5.5).

Chronos (foundation model)

Chronos-T5-small (Ansari et al., 2024) is a transformer that was pre-trained on a large and diverse dataset of time series, and was used in the zero-shot setting without any training or fine-tuning on the Chronos-T5-small test set. The model was provided with the 512 hours of previous Appliances data as context for each fold and required to generate a probabilistic prediction for the next 24 hours of data; the median of 20 sampled forecasts was taken as the point forecast.

Results

Overall comparison

Table 1 reports the mean metrics across all 14 folds for all 8 models, ranked by RMSE.

Table 1: Final model comparison (mean across 14 rolling-origin folds)

Rank	Model	RMSE (%)	MAE (%)	MAPE (%)
1	SARIMAX(2,1,6) $\times$ (0,1,1,24)	56.70	37.59	34.85
2	XGBoost (direct multi-horizon)	61.02	44.04	43.73
3	Chronos-T5-small (zero-shot)	61.71	36.02	23.26
4	Mean	66.91	50.26	53.70
5	Weekly Seasonal Naive	69.23	43.46	37.35
6	Daily Seasonal Naive	76.30	48.31	43.33
7	Naive	97.77	85.55	112.91
8	Drift	97.98	85.80	113.31

Benchmarks

Among the five seasonal/naïve benchmarks, Weekly Seasonal Naïve was the strongest (RMSE 69.23, MAE 43.46, MAPE 37.35%) followed by Daily Seasonal Naïve, and Naïve and Drift were the weakest (RMSE $\approx$ 98). This suggests that there is indeed weekly cycle structure to the series beyond just the fact that there are daily cycles (as seen in the household activities), and that the series is not a random walk (Naïve/Drift works so poorly).

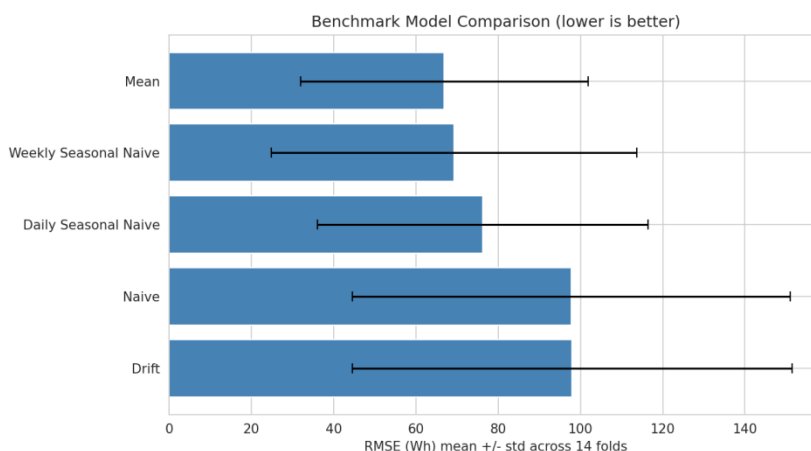


Figure 4: Benchmark comparison by mean RMSE across 14 folds

SARIMAX

The best model (AIC 5101.43) selected by the seasonal AIC grid search was SARIMAX(2,1,6)×(1,0,1,24), which indicates that the daily seasonal term is useful for adding meaningfully to the fit over non-seasonal ARIMA alternatives. The model that performed best was SARIMAX, with an RMSE of 56.70, which was 18% lower than the best model. The ACF of the residuals (Figure 5) has no significant remaining autocorrelation (Ljung-Box Q(1) = 1.06, $p \approx 0.30$), and the ACF is nearly flat. However, the residuals seem to be visually heavy-tailed, suggesting that the model captures the usual hours well but fails to capture larger usage bursts.

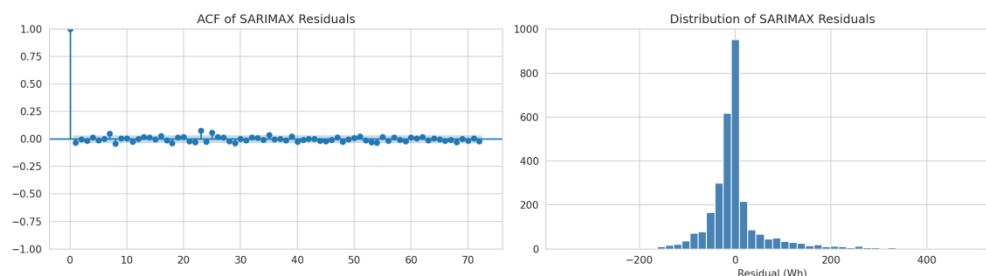


Figure 5: ACF and distribution of SARIMAX residuals, no significant remaining autocorrelation, but heavy-tailed residuals

Figure 6 plots this directly on a held-out fold, which contains a sharp spike around ~ 223 Wh, and SARIMAX does a very good job of following the low, stable overnight time period, with its 95% CI definitely and noticeably containing the true value around the time of the two spikes, even though it misses the value entirely at both times.

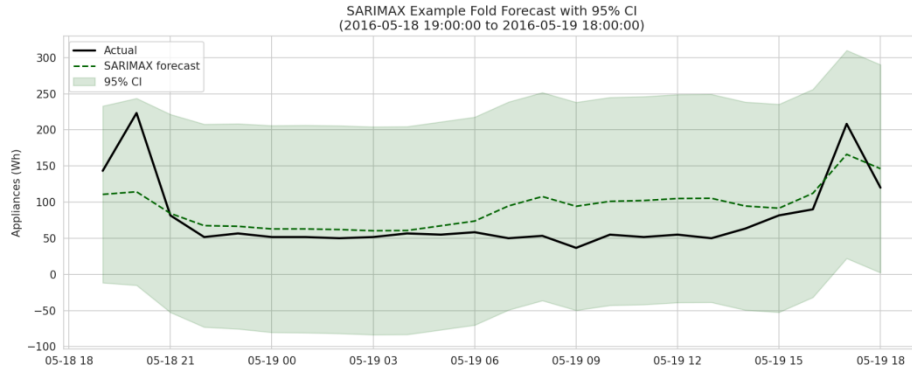


Figure 6: SARIMAX 24-hour forecast vs. actual data collected after the forecast origin, with 95% CI

XGBoost

XGBoost also outperformed significantly over Naive, Drift, and Daily Seasonal Naive, demonstrating that the engineered features are indeed predictive, but did not outperform SARIMAX on any measure, and only beat Weekly Seasonal Naive on RMSE. A very clear pattern emerged for feature importance (Figure 7): calendar/time-of-day features, such as hour, hour_sin/cos, day-of-week, and month, top the ranking, ahead of all raw lag features, and the first non-calendar feature to appear in the ranking is the 168-hour rolling mean. All 27 features are ranked near the bottom of raw lags, which are lag_1 to lag_168.

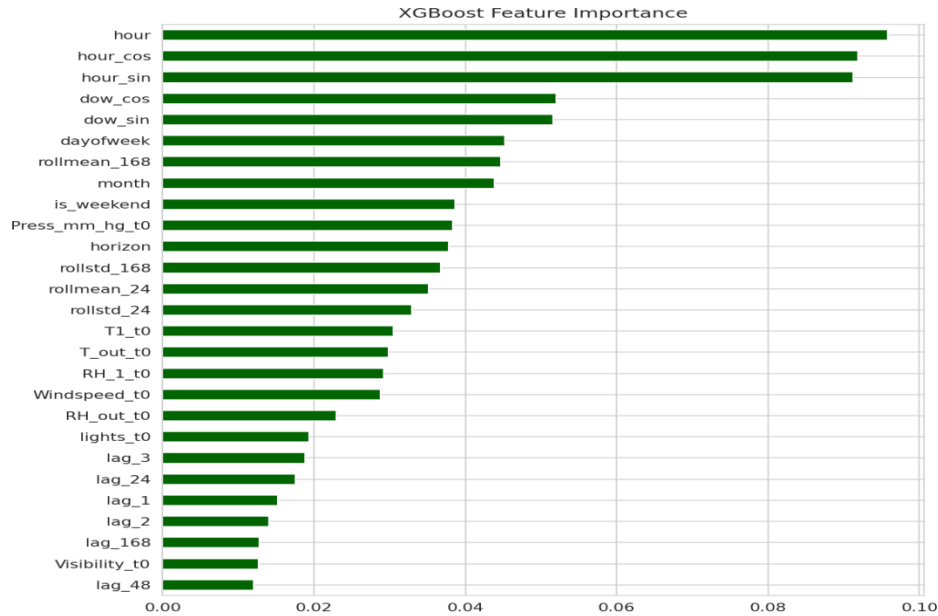


Figure 7: XGBoost feature importance calendar features dominate over raw lags

This is because of the direct multi-horizon design: The same features must be used across all horizons 1 to 24, so the features 'what time of day is the forecast' and 'what time of week is the forecast' are informative at all horizons. In contrast, the feature 'lag_1' is highly informative at horizon 1 and uninformative at horizon 24. This is also the reason XGBoost does not outperform SARIMAX: SARIMAX takes advantage of short-lag autocorrelation at every step as part of its state-space recursion, whereas XGBoost's design does not use that correlation at every step, instead averaging across all 24 horizons.

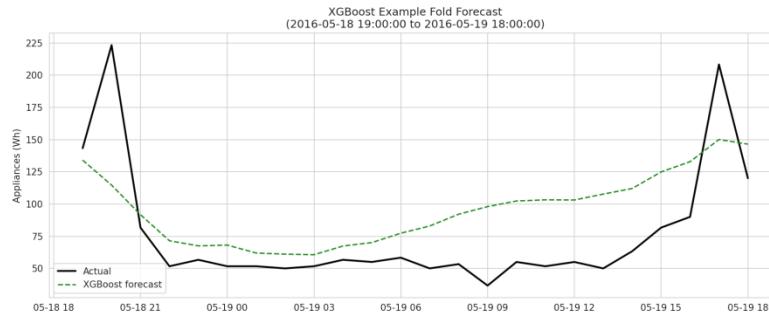


Figure 8: XGBoost 24-hour forecast vs. actual data on the same fold

Chronos (zero-shot foundation model)

Despite no training on this dataset whatsoever, Chronos outperformed all eight models by a very large margin on MAE (36.02%) and by just over a margin on MAPE (23.26% vs 34.85% for the next best model), and was close but slightly behind SARIMAX on RMSE (61.71%). Its relatively high RMSE standard deviation value (42.97%, which is the highest among all the models) reveals the explanation: Chronos is very accurate on typical hours, and the occasional larger miss will raise the RMSE value, but will not equally raise MAE or MAPE.

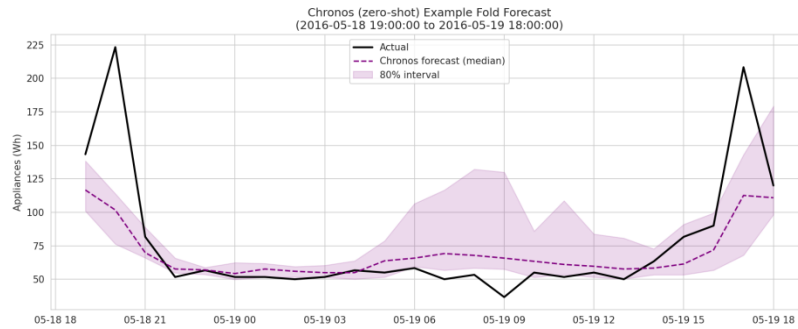


Figure 9: Chronos zero-shot 24-hour forecast vs. actual data, with 80% predictive interval

On the same fold shown in Figures 6, 8, and 9, Chronos tracks the low nightly baseline in addition to SARIMAX and its 80% interval, which, although narrower than SARIMAX's 95% CI, covers most of the true trend, only being significantly exceeded during the two steep spikes, which are not fully captured by any model in this study.

Cross-model comparison

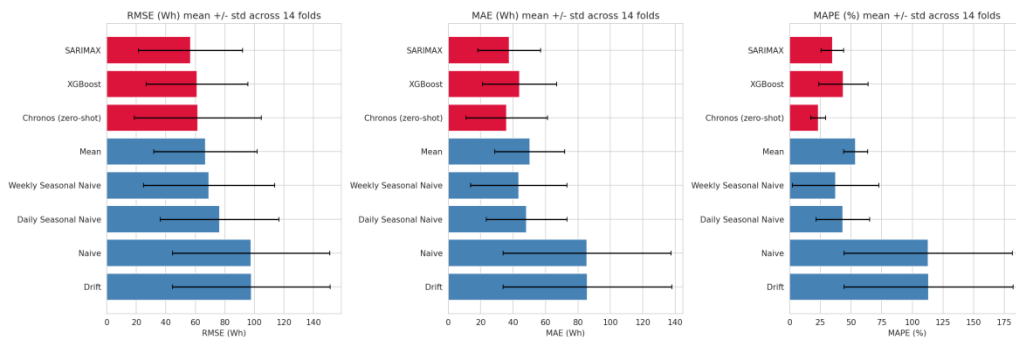


Figure 10: RMSE, MAE, and MAPE compared across all 8 models (red = fitted/pretrained models, blue = simple benchmarks).

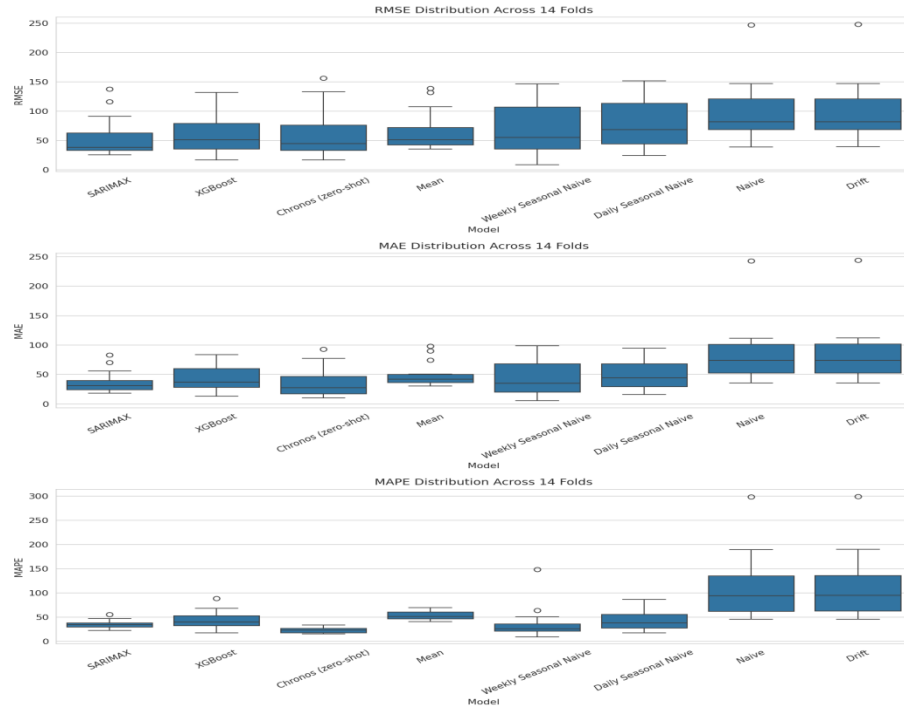


Figure 11: Per-fold error distributions across all 8 models showing consistency, not just average accuracy

Figure 11 adds an important nuance beyond the mean metrics in Table 1: SARIMAX's box is narrow and consistent, explaining its RMSE win, while Chronos's MAPE box is both the lowest and one of the tightest of any model the most reliable typical-case performer fold after fold, not just on average.

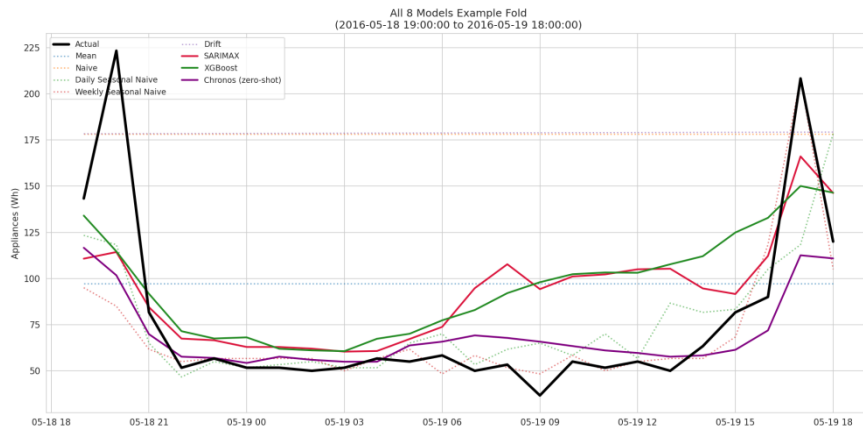


Figure 12: All 8 models' 24-hour forecasts overlaid against the actual held-out data for one representative fold

Figure 12 encapsulates the main results of this study visually: All models fail to capture the two sharp peaks of this fold, though SARIMAX and Weekly Seasonal Naive respond quickly to the increase in usage, whereas XGBoost is less responsive to the rise in the second peak and Chronos is also slow to catch the increase in the usage in the second peak, though smooth and well-calibrated during the flat periods.

Discussion and Critical Analysis

Model-level discussion

The results show a true trade-off between the models, and not just one 'best' model: although SARIMAX has the most efficient, well-documented classical structure and completely disregards the exogenous information, it performs well in situations where the model fits the data; XGBoost, the model with the most features, is not as effective as its competitors in the cases where it accurately predicts the true value, but is far worse in cases where it misses the true value by a large margin; and Chronos, which has no training on this data, performs best on a typical case basis, at the expense of much larger misses when the model doesn't fit the data. All four categories of models in Figures 6, 8, 9, and 12 perform well on the stable periods, and the four models share a common characteristic: they do not fully predict the usage peaks that occur during the series, as is shown by the heavy-tailed residuals in the SARIMAX model (Section 4.3) and as is expected of a series of near-random usage bursts.

Answers to the Six Analysis Questions

Q1. Which benchmark model is strongest, and what does this tell you about the structure of appliance energy use? The weekly Seasonal Naive is clearly better than the other three, with significantly lower values of RMSE, MAE, and MAPE (69.23, 43.46, 37.35%) compared to Daily Seasonal Naive and Naive/Drift (RMSE around 98). This illustrates that the weekly cycle is not just a repetition of the daily cycle and has significant differences across days of the week, and is not a random walk.

Q2. Does SARIMAX improve on the strongest seasonal benchmark? Are daily seasonality, autocorrelation, and exogenous variables adequately captured? Yes, The performance of SARIMAX is superior to Weekly Seasonal Naive across all three metrics: 18% reduction in RMSE, 14% reduction in MAE, and 7% reduction in MAPE. The appearance of an explicit daily seasonal term, in addition to the ARMA structure alone, is an advantage of the model selected, SARIMAX(2,1,6)×(1,0,1,24) (AIC 5101.43), and short-lag autocorrelation is well modelled (Ljung-Box $Q(1) = 1.06$, $p \approx 0.30$, no significant residual autocorrelation). This SARIMAX model, however, is purely univariate, with a noted limitation being that exogenous variables are not captured at all.

Q3. Does the feature-based model improve with lag, rolling, calendar, and sensor features, and which feature groups are most useful? Partially XGBoost significantly outperforms Naive, Drift, and Daily Seasonal Naive but not SARIMAX. The features in general are mostly calendar/time-of-day features, much more important than all the raw lag features, with the 168h rolling mean being the first non-calendar feature to show up in the top ranks (*Figure 7*). This is the direct multi-horizon design: One from horizon 1 is informative from horizon 24, one from horizon 24 is informative from horizon 24, and so on.

Q4. Does the foundation model outperform the simpler, SARIMAX, and feature-based models and is any improvement large enough to justify its complexity? Yes, on typical-case accuracy: Chronos' MAE is 36.02%, which is by far the lowest, and its MAPE is 23.26%, which is significantly lower than the next-lowest, 34.85%, for SARIMAX; its RMSE, 61.71%, is slightly higher than the lowest (61.26%) of SARIMAX. It's the same data, no feature engineering, no tuning, and a significant improvement, so it is worth the added complexity, especially since it is at inference time, not at development time.

Q5. Which variables are genuinely known at the forecast origin? Would using future weather from the test set make this a true or a conditional forecast? Only calendar features (hour, day-of-week, month, and their cyclical encodings) are genuinely known in advance for any future timestamp. Weather/sensor readings are known only up to t_0 . Using the test set's actual future weather values as inputs rather than only current (t_0) readings, as done throughout this study would make the exercise a conditional forecast (conditioned on

already knowing the future weather), not a true autonomous forecast; a real deployment would need genuine external weather forecasts instead.

Q6. Based on accuracy, interpretability, uncertainty, computational cost, and ease of deployment, which model would you recommend for practical smart-home energy forecasting? See Table 2 below. Recommendation: Chronos (zero-shot) despite its relatively low interpretability, it is deployed in several households due to its highest typical-case accuracy and its integrated uncertainty quantiles, without the need for per-household training/cold-start. Where reliability or auditability is more important, and the worst-case scenario is more important (e.g., grid capacity planning), then SARIMAX is the better option as it has the best RMSE and clear structure and can be diagnosed. XGBoost is the least defensible of the three fitted models here: The three most expensive models to engineer, and it doesn't clearly outperform either of the other two.

Table 2: Model trade-offs for practical deployment (Q6)

Criterion	SARIMAX	XGBoost	Chronos (zero-shot)
Accuracy	Best RMSE (56.70)	Middling on all metrics	Best MAE/MAPE, competitive RMSE
Interpretability	High explicit order, diagnostics	Medium feature importances only	Low black-box transformer
Uncertainty	Principled CIs (state-space)	None natively	Principled quantile intervals
Compute cost	Cheap per-household fit	Fast, needs feature pipeline	Zero training; heavier inference
Deployment ease	Cold-start issue (needs history)	Cold-start issue + pipeline upkeep	No cold-start works immediately

Future Improvements

1. Add exogenous regressors to the SARIMAX (true SARIMAX model) using current weather/sensor readings, and test whether classical structure + covariates can bridge the gap with Chronos in terms of MAPE.
2. Train per-horizon specialist models rather than one direct multi-horizon XGBoost to see if the short-lag signal that is lost in the current design (Section 4.4) is restored, and tune hyperparameters using cross-validated search.
3. Test larger variants of Chronos or other foundation models like TimesFM (Das et al., 2024) and calculate the accuracy/inference-cost difference between CPU and GPU.
4. Incorporate conformal prediction in XGBoost for native uncertainty intervals and expand with longer historical record, as zero-shot foundation models are likely to have a relative advantage precisely when task-specific data is limited.

References

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